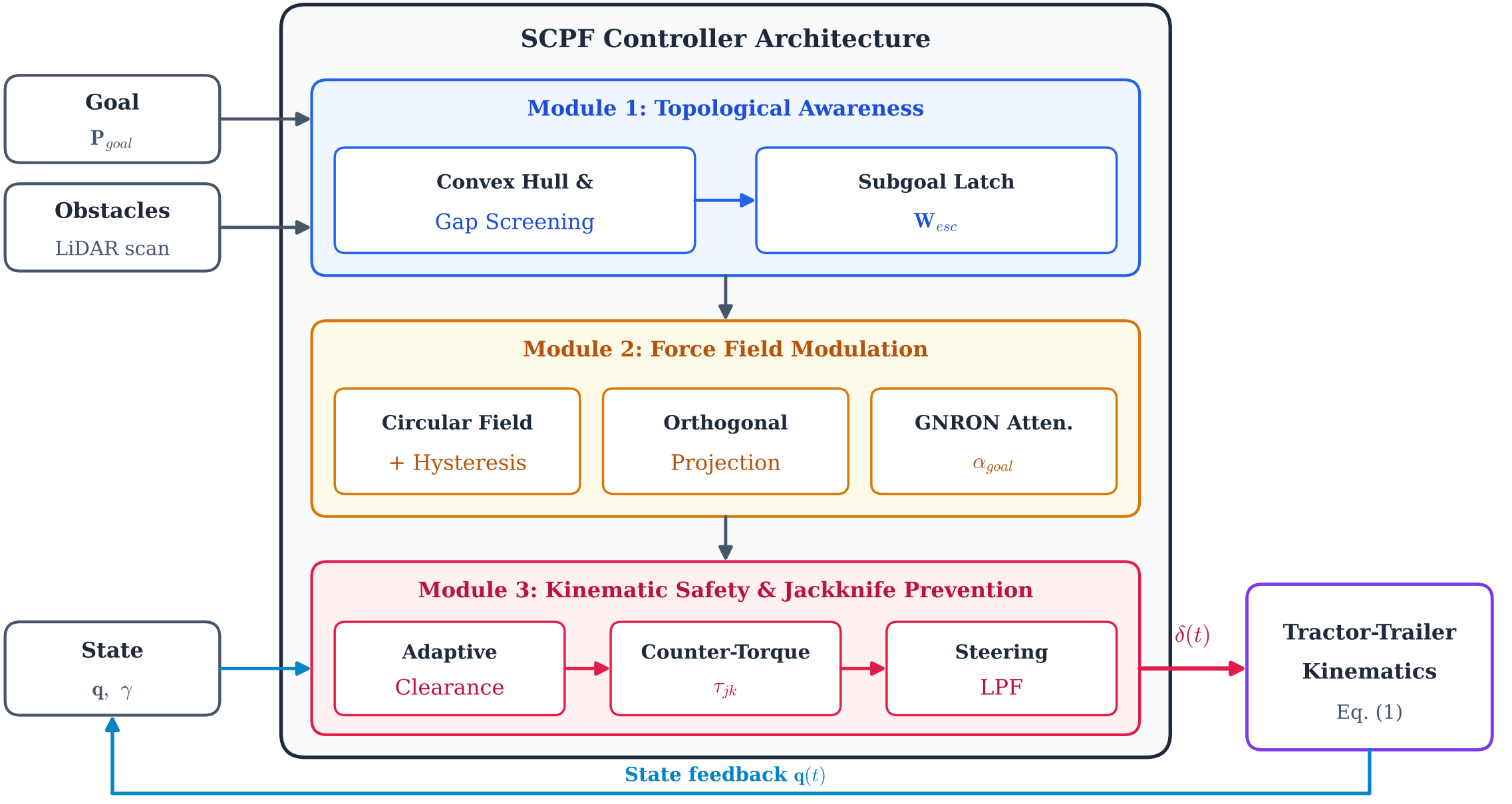
SCPF Controller Architecture
Goal
$\mathbf{P}_{goal}$
Obstacles
LiDAR scan
Module 1: Topological Awareness
Convex Hull &
Gap Screening
Subgoal Latch
$\mathbf{W}_{esc}$
Module 2: Force Field Modulation
Circular Field
+ Hysteresis
Orthogonal
Projection
GNRON Atten.
$\alpha_{goal}$
Module 3: Kinematic Safety & Jackknife Prevention
Adaptive
Clearance
Counter-Torque
$\tau_{jk}$
Steering
LPF
State
$\mathbf{q},\ \gamma$
$\delta(t)$
Tractor-Trailer
Kinematics
Eq. (1)
State feedback $\mathbf{q}(t)$